



# NUCLEAR POWER CORPORATION OF INDIA LIMITED

WELDING PROCEDURE SPECIFICATION (WPS) - ASME QW-482

| SECTION 1 - IDENTIFICATION |               |                      |                             |
|----------------------------|---------------|----------------------|-----------------------------|
| WPS NO.                    | NPCIL-GT-1-43 | REVISION NO.         | R-1                         |
| DATE                       | 2014-07-07    | QUALIFICATION STATUS | Qualified up to 4.5 mm only |
| SUPPORTING PQR NO.(S)      | L             |                      |                             |

| SECTION 2 - WELDING PROCESSES |                 |                        |                        |
|-------------------------------|-----------------|------------------------|------------------------|
| WELDING PROCESS (ROOT)        | GTAW            | WELDING PROCESS (FILL) | GTAW                   |
| TYPE                          | Manual          | JOINT DESIGN           | Groove                 |
| BACKING                       | With or Without | BACKING MATERIAL       | As per design document |

| SECTION 4 - BASE METALS (QW-403) |                                          |                          |               |
|----------------------------------|------------------------------------------|--------------------------|---------------|
| P-NO.                            | P-No. 1 Group 1/2                        | TO P-NO.                 | P-No. 43      |
| MATERIAL SPEC / GRADE            | Synthetic but code-style material family |                          |               |
| THICKNESS RANGE (GROOVE)         | 1.5 mm to 19 mm                          | THICKNESS RANGE (FILLET) | All Thickness |

| SECTION 5 - POSITIONS (QW-405) |     |                     |                 |
|--------------------------------|-----|---------------------|-----------------|
| POSITION(S) OF GROOVE          | All | WELDING PROGRESSION | Vertical Uphill |
| POSITION(S) OF FILLET          | All | PWHT                | Not Required    |

PREPARED BY

REVIEWED BY (QA)

APPROVED BY



| SECTION 6 - PREHEAT & PWHT (QW-406 / QW-407) |              |                     |          |
|----------------------------------------------|--------------|---------------------|----------|
| PREHEAT TEMP, MIN                            | 10.0 °C      | INTERPASS TEMP, MAX | 250.0 °C |
| PWHT REQUIREMENTS                            | Not Required |                     |          |

| SECTION 8 - ELECTRICAL CHARACTERISTICS (QW-409) |              |
|-------------------------------------------------|--------------|
| CURRENT / POLARITY                              | DC-EN        |
| TRAVEL SPEED RANGE                              | 40-80 mm/min |

| SECTION 9 - TECHNIQUE (QW-410) |                              |
|--------------------------------|------------------------------|
| STRING OR WEAVE BEAD           | Root: Stringer / Fill: Weave |
| CLEANING METHOD                | Brushing or Grinding         |
| PEENING ALLOWED                | False                        |

| SECTION 10 - FILLER METALS (QW-404) |        |                    |          |
|-------------------------------------|--------|--------------------|----------|
| SFA SPECIFICATION                   | 5.14   | AWS CLASSIFICATION | ERNiCr-3 |
| F-NO.                               | varies | A-NO.              | varies   |

NOTES & RESTRICTIONS:

Synthetic WPS record used for contrast and edge cases.

JOINT SCHEMATIC

